



Using Video Modeling to Increase Variation in the Conversation of Children with Autism

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The present study assessed the effects of video modeling on acquisition and generalization of variation in the conversational speech of two boys with autism. A video was made showing several versions of several topics of conversation, thus providing multiple exemplars of each conversation. Video modeling consisted of showing each child a video depicting the various versions of the target conversation. A multiple baseline design across children and within child was used. Generalization across setting, peer with autism, typically developing peer, sibling at home, and topics of conversation were assessed. Ancillary measures of question-asking and other social behaviors were taken in free play sessions. The results indicated that both subjects increased variation in their conversation as a result of video modeling, as measured both within conversation and cumulatively across conversations.

Video modeling has been investigated for more than two decades as an effective way to teach a variety of behaviors to persons with autism spectrum disorder (ASD). This includes play skills, functional living skills, social behaviors, academic skills, and conversational skills (Apple, Billingsley, & Schwartz, 2005; Charlop-Christy, Le, & Freeman, 2000; Charlop & Milstein, 1989; Nikopoulos & Keenan, 2003; 2004). Few video modeling studies to date have addressed social communication skills (Bellini & Akullian, 2007). Moreover, only two studies have targeted the social communication skills of conversational speech (Charlop & Milstein, 1989; Sherer et al., 2001).

Unlike other skills within the social communication domain (i.e., social initiations, verbal responses, and compliments), conversational speech requires the use of multiple complex language skills such as initiation and expansion of a conversational topic, establishing an interactive "to-and-fro" pattern of conversation, and fi-

nally, maintaining a verbal exchange (Charlop-Christy & Kelso, 2003). Although some of the video modeling studies have examined aspects of conversational speech, few have targeted the entirety of the behavioral chain required for conversation.

Rather, perhaps because of its complexity, conversational speech for persons with ASD has primarily been targeted through the use of scripts and cue cards (Charlop & Milstein, 1989; Charlop-Christy & Kelso, 2003; Goldstein & Cisar, 1992; Krantz & McClannahan, 1993), which allow the chain of responses necessary for a conversation to be broken down into discrete lines, thereby simplifying it.

In one such study, Krantz and McClannahan (1993) implemented written scripts to facilitate social initiations in four children with autism. During a structured art project, participants were presented with scripts and were instructed to read initiation phrases to typically de-



veloping peers in their proximity. The scripts were then systematically faded from end to beginning, until the scripts were no longer present. Results showed that when the scripts were introduced, peer initiations increased, and as the scripts were faded, unscripted initiations increased. While this research demonstrates the efficacy of using written scripts to teach social initiations to persons with ASD, initiations are only one part of the behavioral chain required to engage in conversational speech. Further, this procedure is limited to individuals who can read and carry out corresponding comprehension.

Charlop-Christy and Kelso (2003) further investigated the effectiveness of a cue card/written script program in specifically teaching conversational speech to children with autism. Conversations were scripted and then broken up into lines that were written on cue cards. The conversant would say her line of the conversation and then present the cue card for the child with autism to read and respond. The conversant would subsequently reply with her next line of conversation and present the next cue card to the child to read and respond with, and so on. The presentation of the cue cards was gradually faded out until the conversant and the child engaged in the short conversation without assistance. Generalization without the cue cards across people, setting, and an untrained topic occurred. Importantly, for two of the three participants, response generalization occurred, such that after one conversation was taught using the cue cards, the children began engaging in appropriate conversational responses across other conversations (untrained scripts). Once again, however, this procedure may only be used with children who can read and comprehend what they are reading.

Although these studies show the potential of cue cards and written scripts for teaching conversational skills, the applicability of these procedures is limited. For example, as mentioned, these teaching methods require participants to be able to read and comprehend. The limitation of literacy also may reduce the age range of children who can benefit from these techniques. Furthermore, while the assumption that children with ASD would be on the higher functioning end of the spectrum because they would have the prerequisite verbal skills, many children with autism are able to engage in verbal exchanges but are not able to read. Another potential limitation is that the use of cue cards and written scripts does not necessarily

teach the nonverbal aspects of a conversational interaction such as eye contact and body positioning.

As mentioned, video modeling has been documented in the literature as being an effective technology to teach a variety of skills to individuals with ASD. It seems feasible that video modeling would be appropriate to teach conversational speech as well (Charlop & Milstein, 1989). Video modeling addresses the limitations of scripts and cues by eliminating the requirements of reading ability, functioning level, and age. Moreover, video modeling is associated with additional advantages such as cost efficiency (Charlop & Milstein, 1989), rapid learning (Charlop-Christy, Le, & Freeman, 2000), promotion of generalization (Matson & Sweezy, 1994), and ability to present an entire conversation without the need of fading.

Two studies have examined the use of video modeling to teach persons with ASD conversational skills (Charlop & Milstein, 1989; Sherer et al., 2001). Sherer and colleagues (2001) investigated whether children with autism could learn a basic conversational exchange via video modeling. In this study, a conversational exchange consisted of an answer response plus posing a question. For example, the conversant may ask, "What is your favorite food," whereas the modeled appropriate response for the child was "My favorite food is pizza. What is your favorite food?" Three out of five participants rapidly met criterion using video modeling, but two participants did not learn the behavior. All the children who met criterion generalized the behavior across persons, settings, and time. It is important to note, albeit not unexpected, that these children were not able to generate new, untrained responses to alternate generalization questions (i.e., response generalization). That is, they could not vary their responses to other topics of conversation.

In the second study, Charlop and Milstein (1989) taught conversational speech to three children with autism using video modeling. Here, conversational speech was defined as the "to-and-fro" verbal exchange between two people sustaining the interaction. The conversations that were scripted and later modeled via video each consisted of seven lines: four for the conversant and three for the child. The conversant typically started or ended the conversation. Each line in the conversation for the child consisted of an answer and a question. This would sustain the conversations. For example, the conversant in the



video asked, "What do you have?" The model replied, "A box. Are you holding something?" The conversant then said, "Yes, a box. What's in your box?" The model responded, "A ball. Is there something in your box?" and so on. Thus, the study required the child to maintain the topic of conversation and engage in multiple verbal exchanges.

All three participants met criterion on the video modeling conversations, and all participants demonstrated generalization across people, setting, and stimuli. For all participants, exposure to the video modeling treatment resulted in appropriate conversational speech across two to four novel conversations. However, only one participant demonstrated generalization to an abstract conversation (response generalization); the other two participants demonstrated partial use of conversational speech to an abstract conversation.

In addition, the study assessed whether video modeling would increase the children's use of appropriate novel responses that varied from the modeled conversation. These ancillary data are important because video modeling has been criticized for resulting in participants just learning a rote way of responding. All participants increased their response variation during the video modeling treatment, and the variation continued to increase during generalization probes. These data are preliminary, however, and additional research is needed to address the use of video modeling as a method to not only teach conversational speech but to simultaneously facilitate response variation within conversational speech.

The present study was designed to expand previous research on using video modeling as a technology to teach conversational speech to children with autism; specifically, to facilitate variation in conversational speech skills. The study had three goals: (a) to systematically assess the effects of video modeling on increasing response variation in conversational speech; (b) to assess the effects of video modeling on generalization of these varied conversational skills across persons, settings, and topics of conversations; and (c) to measure any concomitant changes in the children's social behaviors and their amount of question-asking in unstructured free play sessions.

Method

Participants

Participants were two boys with autism who attended an after-school behavioral treatment program. Both children had been diagnosed with autism spectrum disorder according to the *Diagnostic and Statistical Handbook of Mental Disorders*, IV-TR, #299.00 (American Psychiatric Association [APA], 2000) by licensed psychologists at two independent agencies. Both were verbal and generally answered questions but only occasionally asked questions. When asking questions, they typically asked the same questions repeatedly, which rarely served to maintain conversations. Both boys seldom generalized newly acquired speech skills, or other treatment gains, to non-teaching settings. The two boys were considered to be on the mid to higher end of the autism spectrum due to their speech skills.

Gary was a 9-year-old boy, whose question-asking behavior was primarily rote and repetitive speech. He tended to ask the same few questions over and over, such as "Where are we gonna go?" or "What are we gonna do?" He rarely initiated conversations or spoke spontaneously. The majority of his speech was echolalic or about topics he was preoccupied with, such as Honda cars, nurses, and other unusual topics for a 9-year-old. He engaged in stereotypy of finger rubbing, rocking, and eye squinting.

Connor was an 8-year-old-boy who also had difficulty maintaining conversations and seldom asked questions. He often engaged in inappropriate speech by perseverating on topics with which he was preoccupied, such as clouds and deserts. When asked a question, he generally gave a one- or two-word answer, although his speech about his topics of obsession indicated that he was capable of much longer utterances.

Conversants

During baseline, treatment, testing, and probe conditions the conversant was the second author. During the generalizations probes, the conversant was an experimenter, a participant's sibling, a therapist from the after-school program, or a developmentally typical peer.



Materials

Six sets of toys (e.g., Big Bird and Bert; a monster and a typewriter; a robot and a coloring book; a car and a jet transformer; blue, red, and green yo-yos; and a black and a red yo-yo) were chosen as stimuli to elicit conversation. These toys were determined to be novel (no previous contact) to the children based on parent, teacher, and therapist reports. Additionally, these toys were determined to be of interest to the children based on pre-experiment choice assessments. That is, when individually presented with other toys, the children chose these toys on two out three trials per toy. A variety of conversation scripts about the toys were created.

Conversation scripts. Each toy set served as the basis of a conversation topic (see Table 1). For each conversation topic, seven conversational scripts were written. As depicted in Table 2, for each conversation topic (e.g., robot and typewriter) a set of five scripts was written and identified by a letter and a number. The letter denoted the topic of the conversation (e.g., Script A, robot and typewriter) and the number denoted a particular version or variation of the script (A1, A2, A3, A4, A5). For example, for Conversation A, about the robot and the typewriter, there were five versions, A1, A2, A3, A4 and

A5, so that while any one of these was about the robot and typewriter, they all consisted of different lines so each child could potentially be presented with a number of varying conversation scripts modeled and ideally generate varied responses. Each conversation topic also included two additional scripted conversations: (a) Conversation A-test, which was designed to assess if the child learned conversational speech through video modeling and how much of that speech consisted of varied responses; and (b) Conversation A', which was identical to Conversation A, except that the toys that served as the topic of conversation were changed so that the children had to make modifications in their responses to reflect the ability to speak about the toys that were immediately present and relevant, as opposed to merely memorizing the lines from the scripted conversations that were modeled for them. For example, as seen in Table 1, Gary's Conversation A is about yo-yos. His Conversation A' is about another pair of yo-yos that differed from the original pair in color. Thus, Gary had to change some of his conversation responses so that they were appropriate to this new stimulus set of yo-yos. In other words, he had to talk about new features of these new yo-yos, and these conversation lines had not been modeled for him. This is an assessment of generalization across stimuli.

Each conversation script, as illustrated in Table 2, consisted of four lines for the conversant who initiated the conversation and three lines for the child. Each conversation started with a question for the child and ended with an answer to the child's last question. The middle of the conversation consisted of questions and answers between the conversant and the child. For the three responses available for the child to make, each was considered correct only if the child provided an appropriate answer and follow-up question.

Videos. VHS videotapes were used for video modeling. For each conversation topic (i.e., topics A through D), the experimenter randomly picked and videotaped three versions of conversations (e.g., A1, A2, A3; B1, B3, B5, etc.). [Note: For the remainder of this article, each set of three conversations will be referred to as a separate video. These videos consisted of conversations that were clearly modeled by adults. To facilitate generalization across conversants, many different models were used.]

Table 1

Conversation Topics per Participant

Conversation	Topic
Gary	
A	blue, red, and green yo-yos
A'	black and red yo-yos
B	Big Bird and Bert
C	car and jet transformers
D	robot and coloring book
Connor	
A	robot and coloring book
A'	monster and typewriter
B	Big Bird and Bert
C	car and jet transformers
D	blue, red, and green yo-yos



Table 2

Examples of a Conversation Script Used for Connor (Robot and Coloring Book)

Conversation D1

CONVERSANT: What color is the robot?
 CHILD: Red, blue, yellow, and white. What color are his hands?
 CONVERSANT: Blue. What is he holding?
 CHILD: A coloring book. Does he color?
 CONVERSANT: No. Do you like to color?
 CHILD: Yes. Can we color a picture?
 CONVERSANT: Yeah, great idea!

Conversation D'

CONVERSANT: What color is the monster?
 CHILD: Blue. What color are his hands?
 CONVERSANT: Silver. What is he holding?
 CHILD: A typewriter. Does he type?
 CONVERSANT: No. Do you like to type?
 CHILD: Yes. Can we type?
 CONVERSANT: Yeah, great idea!

Conversation D2

CONVERSANT: What is that toy?
 CHILD: A robot. What's his name?
 CONVERSANT: Ralph the robot. What color are his ears?
 CHILD: Red. Do they move?
 CONVERSANT: Yes, they grow. Do you want to see?
 CHILD: Sure. May I pull his ears?
 CONVERSANT: Sure.

Conversation D3

CONVERSANT: Is the robot holding something?
 CHILD: Yes. What is it?
 CONVERSANT: A coloring book. Do you want to see some pictures?
 CHILD: Yes. Are there pictures of robots?
 CONVERSANT: No. What's in the picture?
 CHILD: A dog. Can I fill in his ears?
 CONVERSANT: Yeah. That would be fun.

Conversation D4

CONVERSANT: Is the robot tall?
 CHILD: Yeah. Does he talk?
 CONVERSANT: No. Is he smiling?
 CHILD: Yes. Do his arms move?
 CONVERSANT: Yeah, watch. Do you want to shake his hands?
 CHILD: Sure. How does he move?
 CONVERSANT: He rolls. Look!

Conversation D5

CONVERSANT: Does Ralph have feet?
 CHILD: No. How does he walk?
 CONVERSANT: He rolls. Does he have hands?
 CHILD: Sure does. Can he pick things up?
 CONVERSANT: Yes. Would you like to see?
 CHILD: Yes. Can he pick up the book?
 CONVERSANT: That's easy.

Corresponding D-Test Conversation

CONVERSANT: Is that a robot?
 CHILD: Yes. Is the robot cute?
 CONVERSANT: Yes. Is the robot big?

Setting

Video modeling sessions were conducted two afternoons per week in one of two (2.9m x 2.9m) therapy rooms at the after-school program the children attended. Each room contained a table, chairs, a cabinet for toys, an intercom for audio access, and a one-way mirror for viewing. These rooms are referred to as the treatment rooms. During the video modeling procedures, a VHS video cassette player, the stimuli, and a television monitor were also located inside the treatment rooms.

Baseline and probe conditions were conducted in the treatment setting and two generalization settings. Generalization settings consisted of a park-like play yard and the participants' home. Conversations with the conversant and probes with a peer with autism were conducted in the treatment rooms. Probes with the child's sibling were conducted at the child's home. Finally, probes with a typically developing peer were conducted at an outdoors play area lawn.



Design

A multiple baseline design across subjects was used. Additionally, a multiple baseline design within subjects across Conversations A and B was used to determine the effects of video modeling on response variation in conversational speech, and a multiple probe design across person, settings, and stimuli was used for Conversations C and D. The order of the presentation of conversations for each participant varied to reduce the likelihood of order effects. Thus, the conversation topic A for Gary was about yo-yos while Conversation A for Connor was about a robot and a coloring book (see Table 1).

Video modeling was presented when the children had completed a predetermined number of A-test and B-test conversations. Gary had five A-test and six B-test conversations, whereas Connor had seven A-test and eight B-test conversations. After video modeling was shown for Conversation A, the children were tested using Conversation A-test. Once criterion had been met on Conversation A-test, probes for generalization were taken.

Procedure

Baseline. The conversant and the child sat facing each other, holding the appropriate stimuli. The conversant began by asking a question about the stimuli (toys) from the prescribed script. The child was expected to answer the question and ask a contextually appropriate question within 10 seconds. If the child did not ask a question, the conversant recorded an incorrect response and continued with the script. If the child answered the original question and asked another question, the conversant responded by answering the child's question and asking another question. This procedure continued until the child had been presented with the opportunity to complete three responses (three lines). The conversation was then closed with the response to the child's last question or with an appropriate ending response.

During the conversation, the toy was given to the child if he asked for it. Thus, the toy served as a natural reinforcer for the child's question-asking. Also, the conversant provided praise such as "uh-huh" or "right" after the child's answer to encourage the child to continue with the conversation. After the conversation was completed, reinforcement, such as tokens, snack, break, or praise, was given for behaviors other than conversing. For ex-

ample, the child was praised for paying attention and sitting nicely while working. However, at no time was the child praised directly for conversing.

Video modeling. Video modeling was presented when the children had completed a predetermined number of conversations with Conversation A scripts.

During video modeling sessions, the experimenter presented three videotaped versions of Conversation A (e.g., A1, A2, A3 or A2, A4, A5). During the presentation of the videotapes, the child sat next to the experimenter facing the television. The child was instructed to watch the television and listen to what the models were talking about. Once the video was finished, the experimenter said, "Now we're going to talk about the toys."

The experimenter then conducted three conversations using Conversation A-test. Prior to video modeling, the amount of response variation across A-test Conversations of each condition was assessed. Thus, variation during baseline for Conversation A, across each A-test with the experimenter, was assessed. If the child demonstrated a greater amount of response variation across the three A-tests than across the A-tests with the experimenter during baseline, the experimenter continued with the A-test conversations.

A-test conversations were conducted until the child had completed as many A-test conversations with the experimenter after video modeling as the child had completed during baseline. Thus, Gary had five A-tests with the experimenter in baseline and five A-tests with the experimenter after video modeling. Likewise, Connor had seven A-tests with the experimenter during baseline and seven A-tests after video modeling.

Criterion was met if the child made three times the number of varied responses than he made during baseline. If the child did not meet criterion on Conversation A, video modeling procedures were repeated. If the child met criterion on Conversation A, he was tested across all of the generalization probes. If the child failed to generalize, video modeling was shown for Conversation B, and the procedure described above was followed. Once the child met criterion, he was again tested for generalization.

Generalization probes. Every probe session was conducted using the same procedures as those described in baseline.



During the probe sessions the conversant was the experimenter, a typically developing peer, or a sibling.

Generalization across persons. These probes were conducted to test for generalization across persons with another child with autism. The conversation was based on the robot and coloring book, which was Conversation D for Gary and Conversation A for Connor (see Table 1). Based on the flip of a coin, Gary was determined to be the initiator of the conversation. To let Gary know that he was to start the conversation, the experimenter said, "Gary, you start and talk to Connor." If he did not begin the conversation, the experimenter instructed him to do so by saying, "Say, is that a robot?" It was expected that Gary might not know how to initiate the conversation since his role, as indicated in the video, was to respond to the conversation upon the initiation. Further, if a child failed to provide a conversational line, he was given a prompt so that the other child would be able to respond and continue the conversation. All prompted lines were scored as incorrect.

Generalization across persons and settings. Two different probes were used to assess generalization across persons and settings together. The experimenter conducted probes with a typically developing peer and the child outside in a park-like play yard. The typically developing peer presented three A-test conversations and three B-test conversations, just as the experimenter did in baseline. The second type of probe, which assessed generalization across persons and settings, was conducted with the child's sibling at home. Both children had siblings who presented three A-test conversations and three B-test conversations, just as the experimenter did in baseline. In both families the child's mother was taught to collect the data.

Generalization across new stimuli. Conversation A' was used to assess generalization to new stimuli. For each Conversation A', the stimuli and the lines used were similar to those in the corresponding A Conversation. For example, for Gary, Conversation A was about a red/blue yo-yo and a green yo-yo. Conversation A' was about a black yo-yo and a red yo-yo. For Connor, Conversation A was based on a robot and a coloring book, whereas Conversation A' was about a monster and a typewriter.

Generalization across topics of conversation. A multiple-baseline design across A- and B-test conversations, and a multiple-probe design across C- and D-test conversa-

tions were used to assess the generalization of conversational skills to topics that had not yet been modeled. After video modeling and after the criterion for Conversation A had been met, the experimenter tested for generalization across the other topics of conversation (Conversations B-D).

Prompted probes. For each conversation (A-D), the experimenter conducted special prompted probes during baseline only for test conversations. The experimenter began the conversation just as in baseline, but if the child was not asking a varied question, the experimenter instructed the child to, "Say something different." These probes were presented prior to video modeling to ascertain that the child could not learn response variation merely by being instructed to do so.

Ancillary Data

Variation in question-asking. The children were videotaped bi-monthly, in a 10-minute free play situation. The child was either with another child with autism, a typically developing peer, or a therapist from the after-school program (these persons were not associated with the study). Transcripts of the children's speech were used to calculate the number of different questions the child asked during these sessions. The number of different questions the child asked during baseline with each of these persons was compared with the number of different questions the child asked these persons following video modeling. This was done to assess the effects of video modeling on the children's use of question-asking in an unstructured social situation.

Social behaviors. The same 10-minute free play situations were scored for the frequency of eight social behaviors (see Table 3). The frequency of these behaviors in baseline was compared with their frequency after video modeling. Social behaviors were assessed to determine if learning variation in conversations had an effect on these social skills.

Dependent Measures

All responses during conversations were initially recorded as correct or incorrect. The criterion for a correct response was providing a contextually appropriate answer and asking a contextually appropriate question per each line of the conversation. The line could be one modeled

**Table 3****Operational Definitions of Social Behaviors**

Social Behavior	Definition
With others	Child is physically within two feet of a person(s).
Parallel play	Child plays with a toy(s) within two feet of another person who is playing in the absence of verbal and nonverbal interactions with others.
Approach	Child walks (moves) within two feet of a person in the absence of a cue from others and stays with the person for 10 seconds or more.
Avoidance	Child walks away (two feet or more) when approached by another person, or resists an interaction (e.g., trying to get out of a hug, or getting off someone's lap).
Interactive play	Child plays with a person by exchanging toys, playing with the same toy(s) at the same time.
Nonverbal response to others	Child provides an appropriate nonverbal behavior in response to a person's verbal or nonverbal behavior while standing within two feet of the individual or while maintaining eye contact.
Nonverbal social initiation	Child begins an activity or affection behavior, within two feet of another person or while maintaining eye contact, in the absence of a cue from others.
Social interaction directed at child	The other person starts an interaction while standing within two feet of the child or orients his/her body toward the child, or maintains eye contact with the child.

the target behavior. So, once the requirement of answering and asking a question was met, and the response was deemed correct, it was determined whether the response met the definition of a varied response.

Variation was determined by the content of the question provided by the child in each line of the conversation. Variation of speech was recorded according to the number of novel appropriate responses the child made in his three lines. Thus, the child had an opportunity to provide three varied responses per conversation. For a response to be recorded as varied, the entire line must meet the definition of a correct response (consist of both an answer and a question) and be of contextually appropriate but different content than any other line previously said in the condition (during baseline or after video modeling). For example, if the child said, "May I play with this?" regarding the blue car in the first line of the conversation, and then asked "May I play with the blue car?" in some later line of a conversation after video modeling, a score of zero variation for the line would be given, because in both instances the child expressed the same content (playing with the blue car). However, if the child asked to "drive" the blue car, he would be credited with two varied responses within the conversation because playing and driving express varied content. If, on his third line, the child, still talking about the blue car, asked to "hold" the car, this too would be considered varied content, and the child's conversation would get a score of 3/3 varied responses. This is an example of a change in content in terms of verb but not object (same blue car). Another example would be if the child asked to play (same verb) with varied objects. Finally, and hopefully, variation in both verb and object would occur (e.g. "The sharks eat pizza"; "I like cheese pizza"; "Sharks really eat fish").

The three lines in the conversations, the within-conversation variation measure, was a content-comparative measure of the three lines. Since variation is a relative concept, there could be two lines that varied from each other or three lines that varied from each other. One line could not vary from another without the other line being counted as varied as well. Thus, there was no way a conversation could have one varied response. The minimum varied responses would be two, and the maximum, three. Therefore, for each conversation, the child could only receive a score of zero, two, or three varied conversational lines. This provides stringent scoring criteria for response variation to avoid inflation of results.

in the video or a novel one. Thus, each child had an opportunity to make three lines or three correct responses per conversation. However, variation in conversation was



Interobserver Reliability

Interobserver agreement was assessed for variation in conversation data, transcripts of free play sessions for question-asking data, and frequency of social behaviors during free play data. This was done for both children in all conditions. Interobserver agreement was determined by calculating the number of agreements divided by the total number of agreements plus disagreements multiplied by 100.

Agreement among observers was calculated as 37% for the conversational data for Gary and 40% for Connor. During the conversations, the second observer either sat behind a one-way mirror, in an unobtrusive location in the experimental room, outside, or at the child's home. For probes taken at the child's home, the mother was trained as the reliability observer. The experimenter compared the number of correct responses and the number of varied responses recorded by each of the observers for the conversations. Interobserver agreement for both children was 100%.

Transcript data were calculated for 41% of all free play sessions across all conditions for Gary and 37% for Connor. Transcripts were assessed using word-for-word agreement. A disagreement was recorded if the primary rater had recorded a word that the secondary rater did not record and vice versa. Interrater agreement for the transcripts was 90% for both children.

Social behaviors during free play sessions across all conditions were determined for 40% of all sessions for Gary and 38% of all sessions for Connor. Observers were trained by reviewing the operational definitions for each social behavior, reviewing and scoring a tape with the experimenter, and then scoring a tape independently to assess reliability. Interobserver agreement was assessed for each behavior during each session, and total agreement was calculated. Interobserver agreement was 96% for Gary and 98% for Connor.

Results

The results of video modeling on conversation variation for Gary and Connor are illustrated in Figures 1-6. Specifically, Figures 1 and 2 show the number of varied responses within conversations for Gary while Figures 3

and 4 show the same data for Connor. Figures 5 and 6 (Gary) and 7 and 8 (Connor) show the cumulative number of varied responses not within conversation, but within condition (baseline, video modeling, post-video modeling) for each conversation across the study.

Overall, the results show an increase in variation in conversational responses for both boys within conversations after video modeling. Additionally, generalization occurred across the probe conditions and to untrained conversations. Cumulative number of varied responses also increased for both boys. Ancillary data suggest an increase in question-asking during free play time as well as increases in a few social behaviors. The results for each child are discussed below.

Within-Conversation Variation

Gary. Gary seldom showed conversational variation across the conditions for Conversation A during baseline (see Figure 1). Two probes were observed when he varied his speech for the A-test Conversation; otherwise, the majority of his data showed no variation during Conversation A and A-test probes during baseline. Prompting ("Say something different") did not facilitate conversation variation. Gary needed only two presentations of video modeling before he met criterion of tripling his variation within conversation compared with baseline. Generalization probes with a typically developing peer and a peer with autism suggested partial generalization.

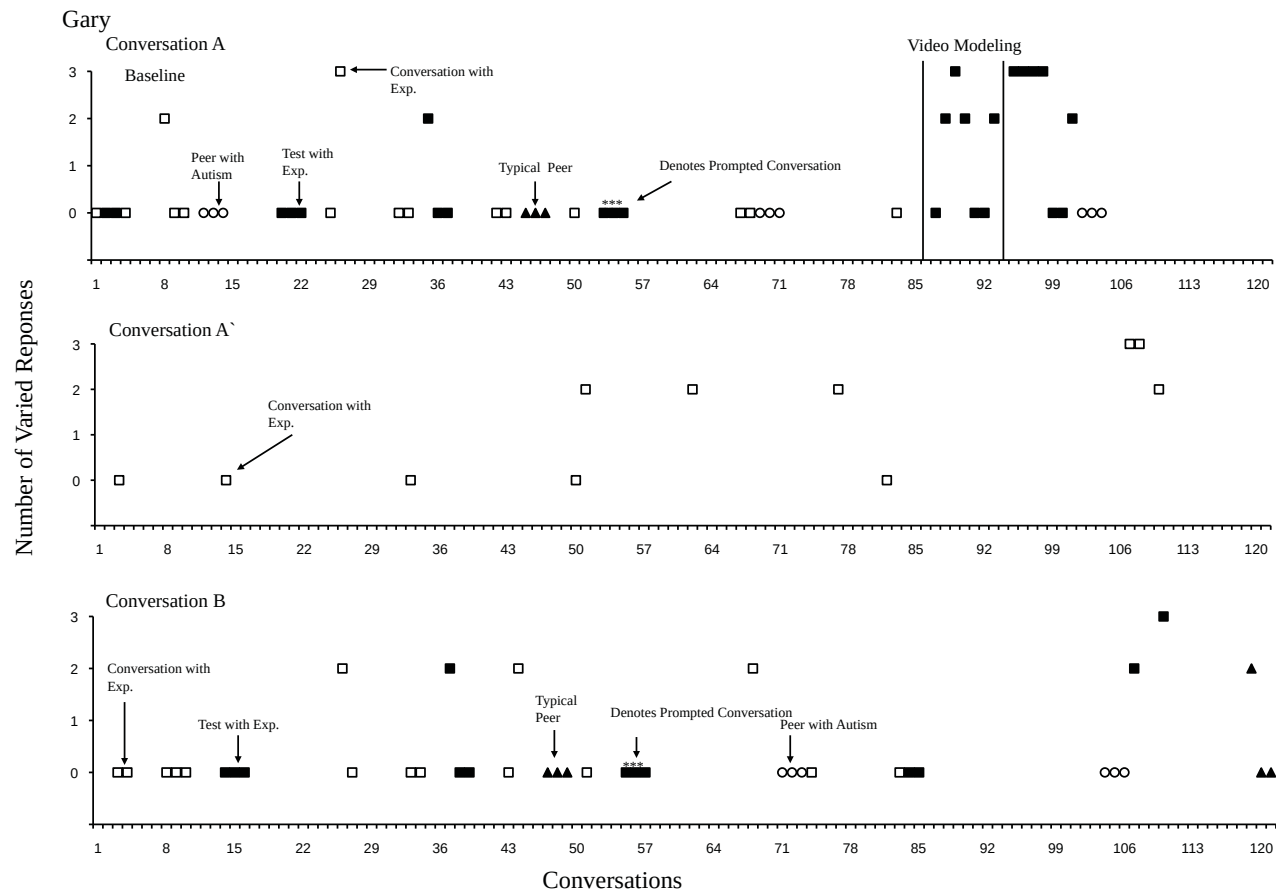
Generalization across stimuli, which was assessed through Conversation A', did not show increased variation after video modeling was presented for Conversation A. While it is unclear whether there is a strong collateral effect of video modeling for Conversation A upon Conversation C, there appears to be an effect for Conversations C and D (see Figure 2).

Connor. Figure 3 shows the effect of video modeling on within-conversation variation for Connor. During baseline, there were few conversations during Conversation A (top panel) in which variation in conversation occurred. However, after two presentations of video modeling, variation in within-conversation increased and met criterion of triple the frequency of baseline. Generalization probe data did not yield positive results for Conversation A. However, there did appear to be improvement in conversation variation for both Conversation A' and



Figure 1

Number of varied responses for Gary within conversations for Conversations A, A', and B with experimenter, peer with autism, and typical peer



Conversation B. Data collected for Conversations C and D show erratic performance (see Figure 4).

Cumulative Variation in Responses Across Conversations

Since there are limits to how increases within conversation in response variation can be measured, it is also important to determine the cumulative amount of varied conversational responses the children learned after video modeling. As seen in Figures 5-8, both Gary and Connor made large increases in the cumulative frequencies of varied and novel responses after video modeling compared with the number of different responses they made during baseline in each condition.

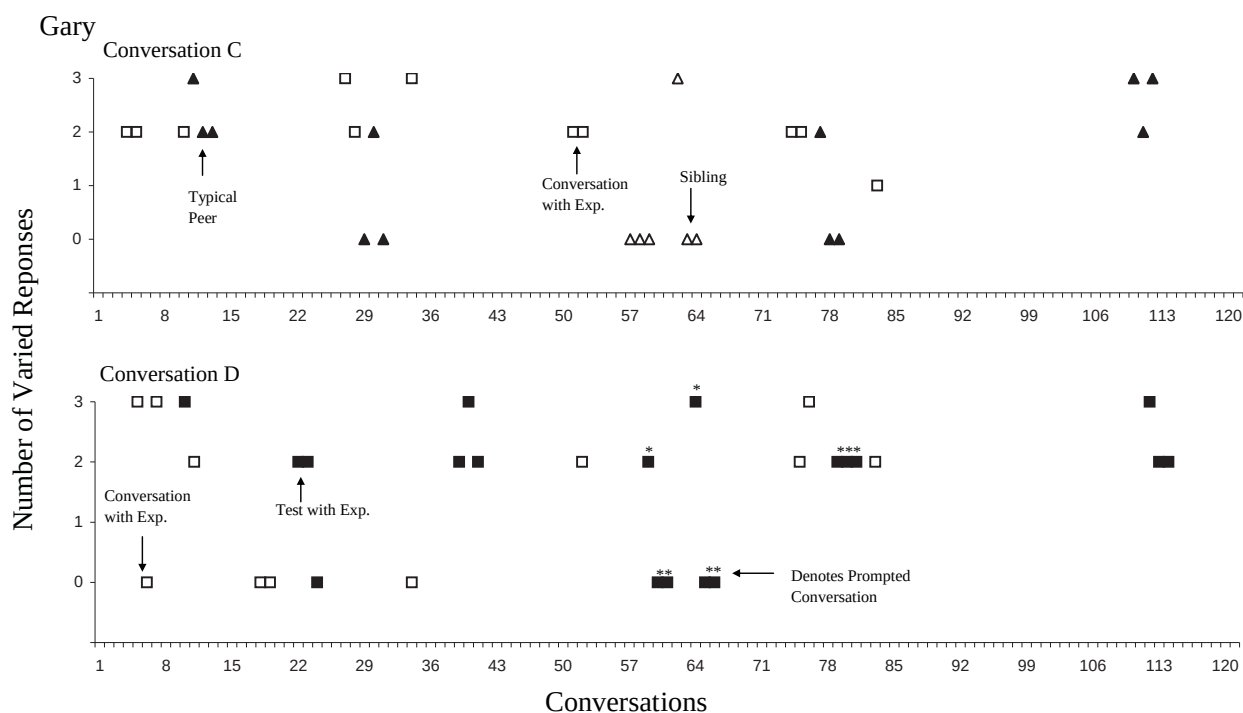
Gary. Gary showed little variation across conversations in each condition during baseline, and after five A-test conversations with the experimenter, he had only asked three varied questions. By comparison, after one presentation of video modeling, he made nine varied responses. When video modeling was presented a second time, Gary asked an additional seven new questions. In addition, he generalized variation across people and settings, and the data showed an increase in variation across topics of conversation (see Figures 5 and 6).

Connor. Connor, showed little variation across conversations in each conditions for Conversations A and B (see Figures 7 and 8). After video modeling was presented, however, he made 8 novel and varied responses and ob-



Figure 2

Number of varied responses for Gary within conversations for Conversations C and D with experimenter, peer with autism, and typical peer



tained a cumulative number of 16 varied responses after the second presentation of video modeling. Generalization occurred across Conversations B-D. However, more varied responses did not occur in the probes for generalization across persons or settings.

Question-Asking During Free Play

The number of varied questions out of the total number of questions asked during the 10-minute free play sessions are listed in Table 4.

Gary. During baseline, Gary asked one question during his free play sessions with a therapist and with a peer with autism, and this question occurred with the therapist. However, since there was no other question to compare it to, this question could not be counted as varied. After video modeling, Gary asked four questions, all in the therapist condition; three of them were varied responses. Thus, Gary increased his variation in question-asking

Table 4

Response Variation During Free Play Sessions

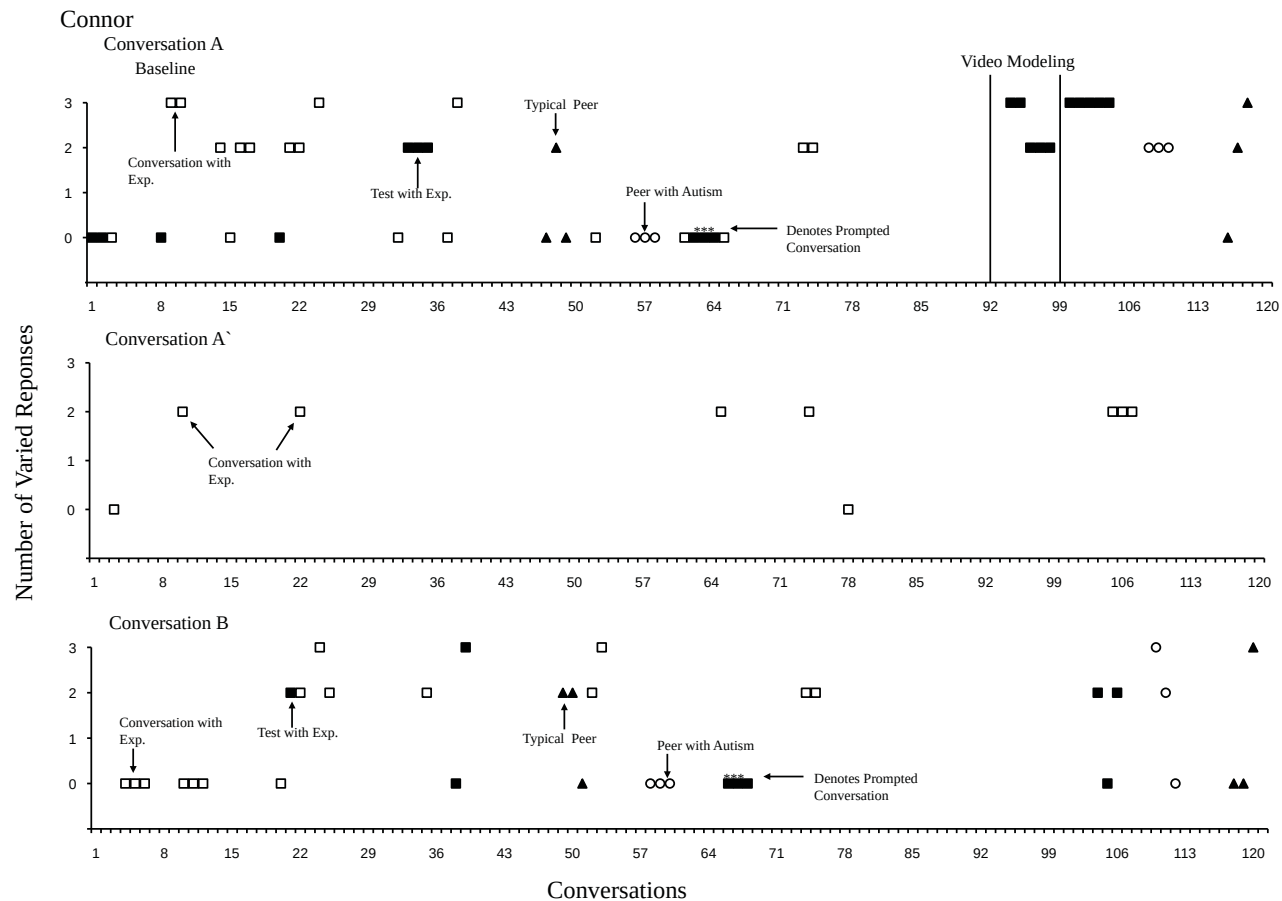
Condition	Baseline	Post-Video Modeling
Gary		
Therapist	0/1	3/4
Peer with Autism	0/0	0/0
Total	0/1	3/4
Connor		
Therapist	3/3	16/18
Peer with Autism	1/2	8/18
Typical Peer	2/2	4/4
Total	6/7	28/40

Note. The scores represent ratios between the number of varied responses and the total number of questions asked.



Figure 3

Number of varied responses for Connor across conversations for Conversations A, A', and B with experimenter, peer with autism, typical peer, and sibling



after video modeling and the frequency of his question-asking increased, but he did not ask a question to his peer, and his overall number of questions was not high.

Connor. Connor displayed more question-asking behavior during baseline than Gary, speaking across all conditions and varying his responses for six out of his seven questions. Importantly, after video modeling, Connor dramatically increased the frequency of his questions to 40 questions, 28 of which were considered varied responses.

Social Behaviors During Free Play

Figures 9 and 10 show the eight social behaviors measured for both children in all conditions for baseline and after video modeling. Changes in the social behaviors did not reveal any consistent findings for the two boys.

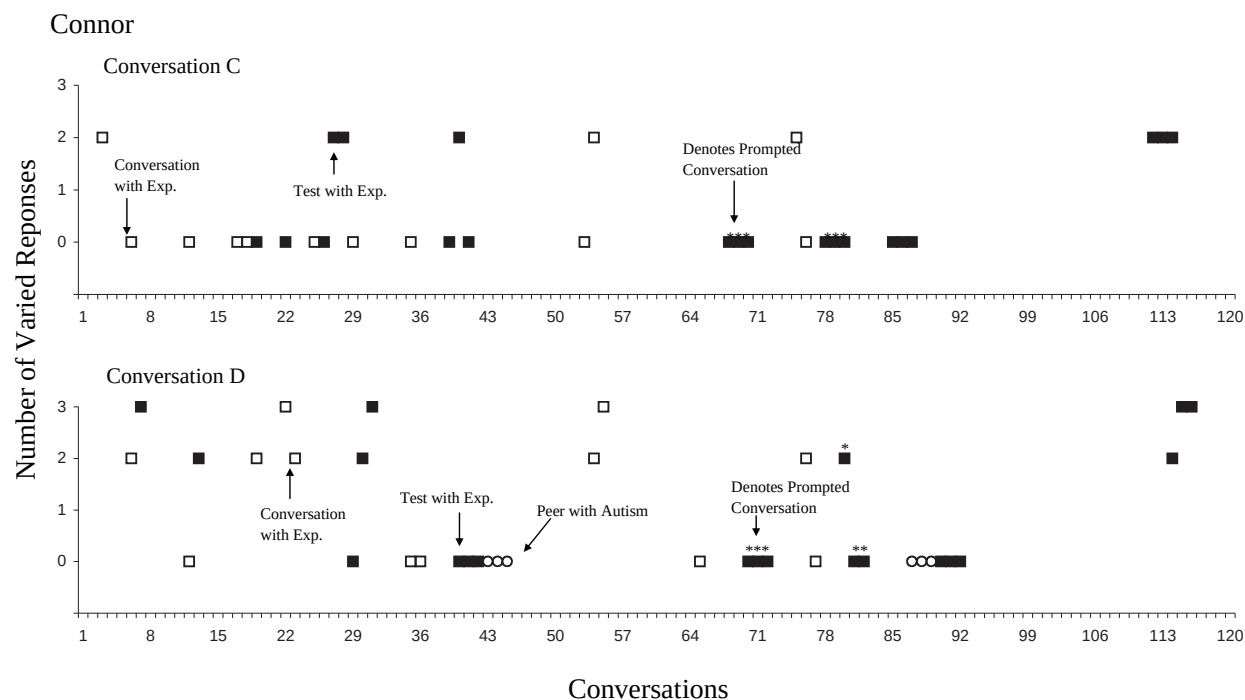
Discussion

In the present study, two boys with autism were taught variation in conversational speech using video modeling technology. Scripts of various topics of conversations were written about a number of toys that were the ba-



Figure 4

Number of varied responses for Connor within conversations for Conversations C and D with experimenter, peer with autism, and typical peer



sis of discussion. For each toy topic, several scripts were written in order to provide examples of response variation. This form of multiple-exemplar training has been shown to increase response generalization (e.g., Charlop-Christy & Daneshvar, 2003; Stokes & Baer, 1977).

The varied conversations were videotaped and presented to the children via a video modeling protocol devised by Charlop and Milstein (1989) and replicated many times in the literature (cf. Charlop-Christy, Le, & Freeman, 2002). Generalization was assessed across untrained but similar toys, settings, typically developing peers, and peers with autism. Additionally, response generalization to other topics of conversations based on different toys was assessed.

The results replicated the previous video modeling study by Charlop and Milstein (1989) showing rapid effects of video modeling on teaching conversational speech after long baselines. However, the purpose of the present

study was to examine whether variation in the children's responding could be increased beyond the modeled phrases depicted in the videos. The results of the present study also show the efficacy of the multiple conversation presentation format as a means to enhance variation in the conversation of the children with autism. Further, generalization probes suggested both stimulus and response generalization.

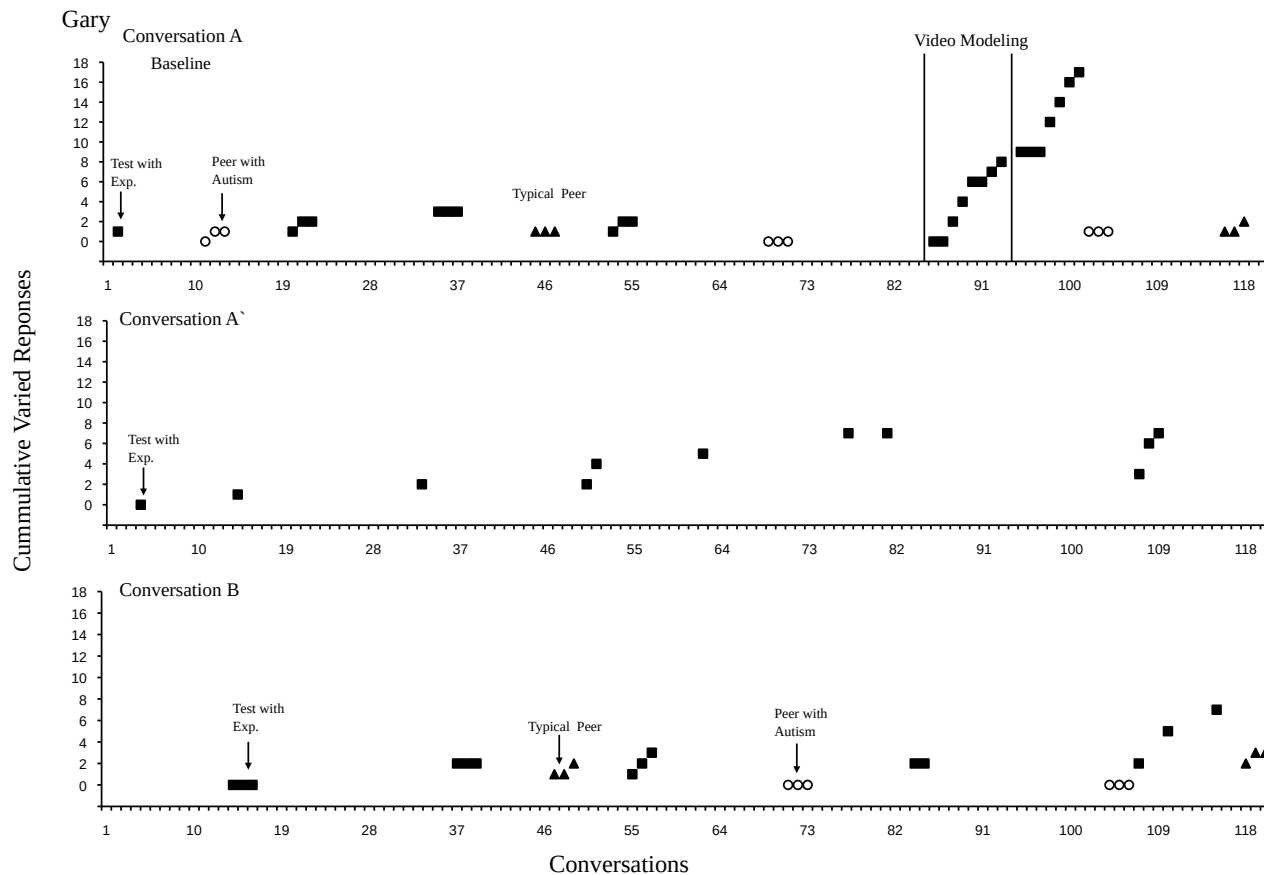
The present research adds to the literature in several ways. First, the findings provide additional evidence for the effectiveness of video modeling in general (Charlop-Christy, et al., 2000) as well as for conversational speech specifically (Charlop & Milstein, 1989; Scherer et al., 2000).

Second, the study shows how a multiple exemplar (multiple conversation) training paradigm (Stokes & Baer, 1977), in this case the presentation of multiple conversations about the same topics in the video, can facilitate



Figure 5

Cumulative variation across test conversations for Gary for Conversations A, A', and B



response variation, response generalization, and stimulus generalization. These results are encouraging in that they demonstrate the effectiveness of a teaching technology, video modeling that has been deemed cost efficient as well as expedient, in teaching a difficult to teach behavior to children with autism. Charlop-Christy and Daneshvar (2003) as well as LeBlanc and colleagues (2003) demonstrated the utility of incorporating multiple exemplars for teaching a perspective-taking task. This may be necessary when teaching more abstract concepts such as conversational variation.

Third, the present research tested the children under naturalistic conditions in which variation in conversational speech would be most desirable; with another peer

with autism, a typically developing peer, and a sibling at home.

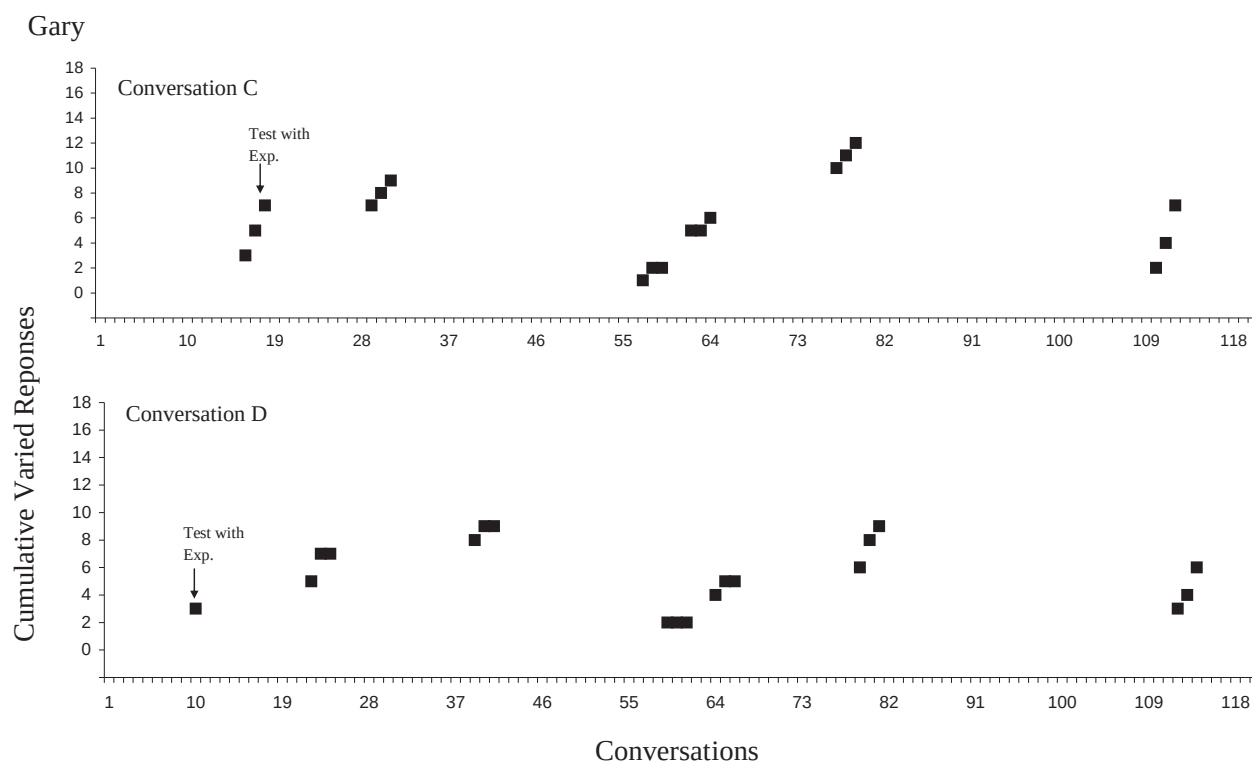
Fourth, the inclusion of ancillary measures of social behaviors allowed for assessment of any concomitant changes that may be seen in association with video modeling. The observations of social behaviors during free play conditions before and after video modeling allowed for such an assessment. Also, the transcription of recorded free play sessions before and after video modeling allowed for a comparison of the number of questions asked before versus after video modeling.

Finally, in addition to the extensive analysis of both stimulus and response generalizations, the dependent variable was measured in two ways. First, we assessed variation



Figure 6

Cumulative variation across test conversations for Gary for Conversations C and D



in conversational speech within conversation. While the data showed positive results, this approach limits what we can assess regarding what the children learned. In the present study, as in previous research (Scherer et al., 2001), the dependent variable had a predetermined number of responses to score. That is, for each conversation there was a maximum of three correct responses. Additionally, due to the nature of content variation, we did not allow for one correct line to be counted as a correct varied response unless there was at least one other response to which its content could be compared. So, our range of within conversation correct varied responses was zero, two, and three responses. While this measure can determine the efficacy of video modeling, and improvement over baseline, it may limit what we can know about what the children learned about response variation as a concept. That is, it prevents us from determining whether the children acquired the “learning to learn” concept (Schreibman, Charlop, & Koegel, 1982)

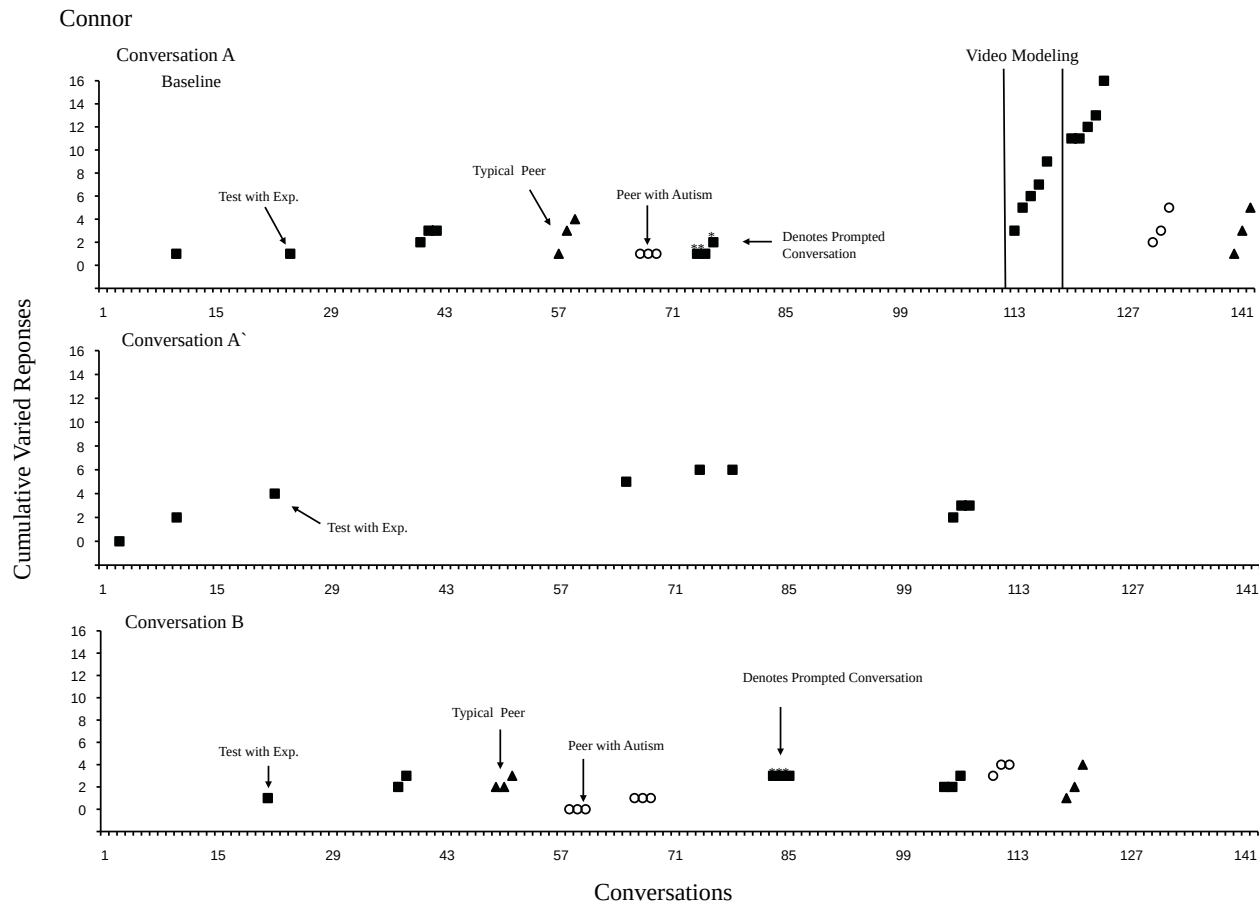
or whether they understood how to take one line in one conversation (say, a line from Conversation C), modify it, and apply it in the context of the toys and topic in Conversation A-test.

This could be measured by the cumulative data figures. From this data analysis, we learned that the children not only learned through video modeling and met criterion in their within conversation variation but they also greatly increased variation in responding after video modeling across the study. From this cumulative analysis, we were also able to anecdotally observe some qualitative changes in the boys’ varied responses. For example, while questions tended to be short and basic during baseline, toward the end of the study, the children’s questions became more sophisticated. For example, Connor asked, “Where is Robinson Crusoe from?”



Figure 7

Cumulative variation across test conversations for Connor for Conversations A, A', and B



The results of the cumulative analysis also provide additional ammunition against arguments that mimicking video presentations of behaviors may merely be imitation and rote learning. While this line of criticism is less applicable here, where we found such variation in response, even within conversation, the cumulative analysis does provide evidence of continual engagement of the desired behavior in incarnations never modeled. Another piece of evidence is offered as well. Recall that when being tested for conversation-variation acquisition, the conversations for each child with autism were initiated by the experimenter. However, when speaking with a peer with autism, one child had to assume the alternate role; that of the experimenter. Gary demonstrated that he was capable of engaging in conversational variation for the

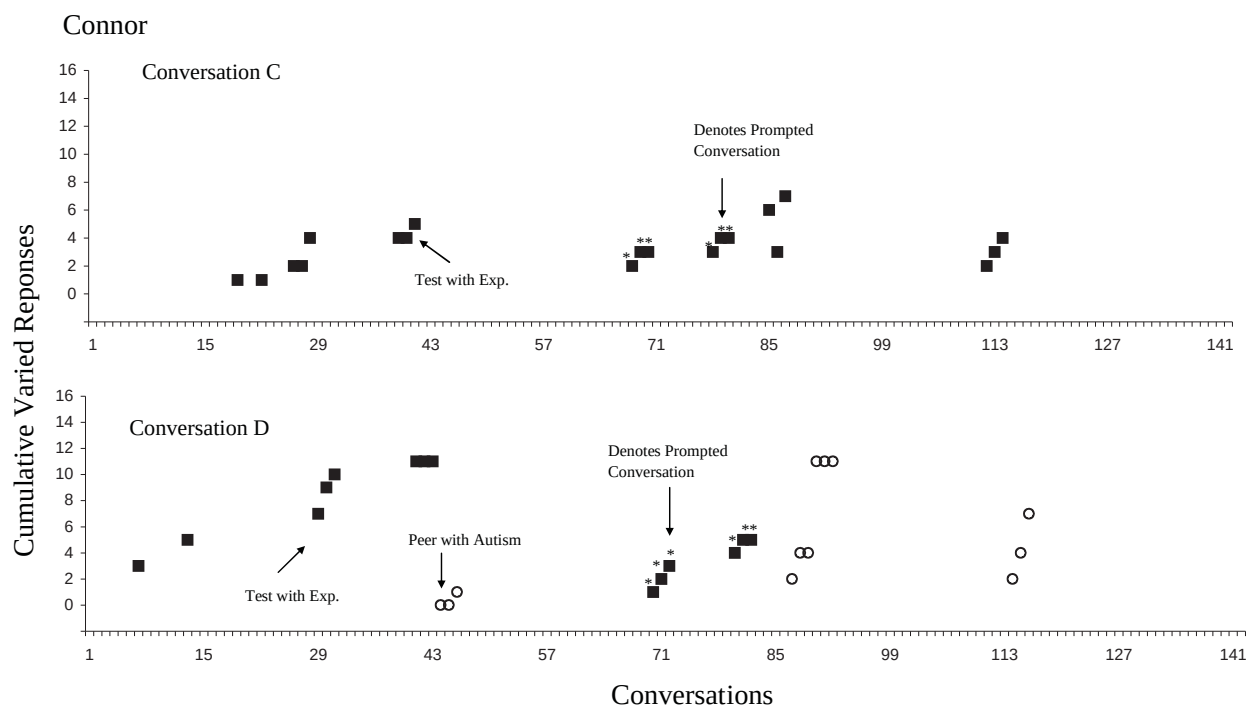
entire conversation, including the experimenter's roles. This, indeed, suggests that variation in conversation as well as varied initiations were learned. Unfortunately, we had only one dyad of children with autism. We felt that to switch off and change the boys around to take turns being initiators might have been too much variation for them. However, it is possible that Connor could have fared just as well as Gary.

Our results support the fast and efficacious findings of video modeling and showed the applicability to conversation variation. However, some of the results show that we need to continue to explore modifications in video modeling to enhance its effects. The generalization effects in the present study were not as positive as



Figure 8

Cumulative variation across test conversations for Connor for Conversations C and D



in other video modeling studies. This may be due to (a) the nature of the target behavior, that being an abstract concept; (b) the nature of autism, that being difficult to obtain generalizable effects; or (c) limitations within our procedures. The inclusion of only two participants suggests the need for replication across more dyads of children. Further, the ancillary data showed mixed results. The free play observations of eight social behaviors did not show anything consistent with the children in the various conditions during baseline and after video modeling. Perhaps variation in conversations has little to no effect upon other social behaviors or perhaps such effects need more time to show up. There are a number of possibilities but there were no immediate effects. However, immediate and dramatic collateral effects were seen with regard to question-asking during free play. Perhaps this is due to the more relevant nature of the behaviors. The number of questions and, more importantly, the variety of questions asked increased after video modeling. This

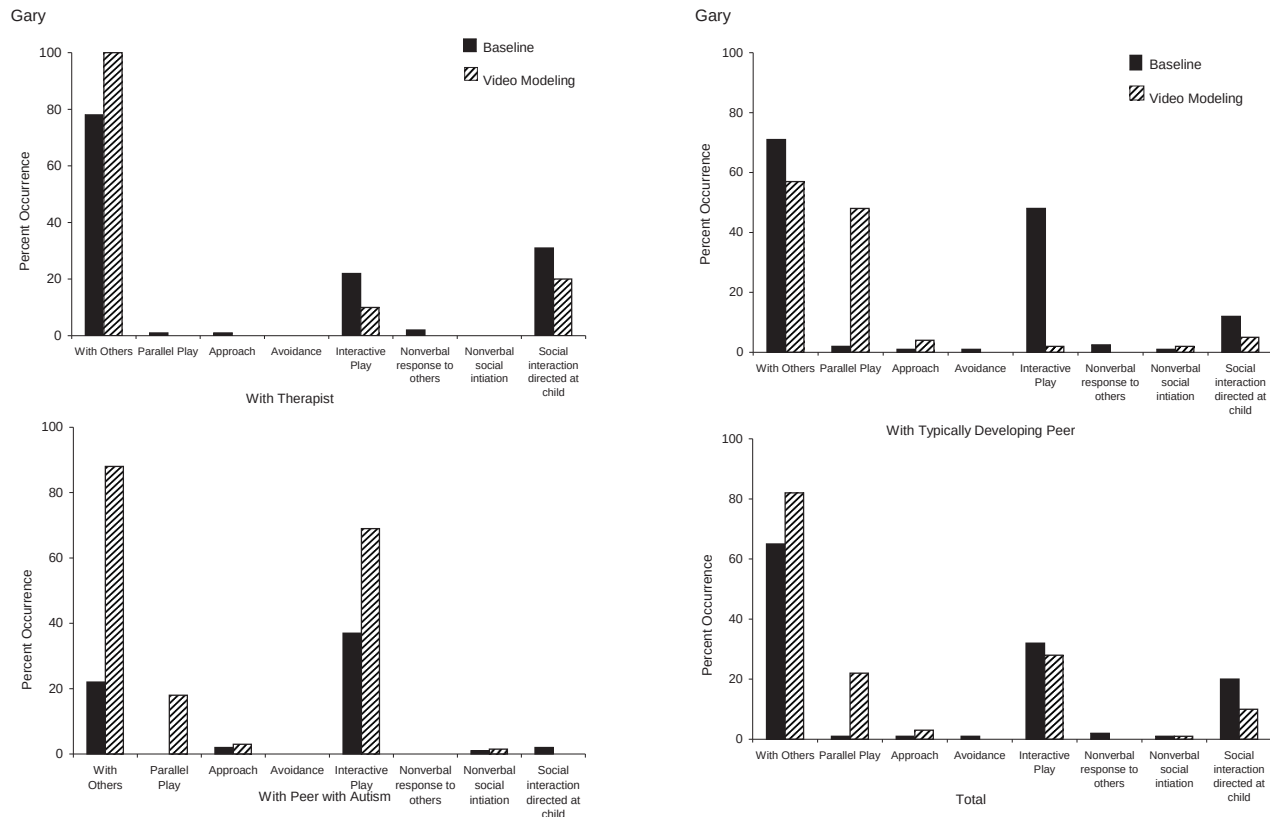
suggests that the varied conversation format might have enhanced the variation of the children's speech in other situations, specifically the number and variation in the questions they asked.

This is an important ancillary gain for several reasons. First, children with autism tend to be responders and not question askers (Schreibman, 2005). Therefore, an important behavior was increased, likely due to the intervention. Once children with autism start asking more questions, they will likely have more interaction with the environment and, therefore, obtain more information. Also, as mentioned earlier, children with autism tend to say the same phrases or ask the same questions repeatedly and perhaps perseveratively (e.g., Kanner, 1943; Rimland, 1964). This is considered one of the inappropriate behaviors of autism (APA, 2000). Thus, with the increase of question variation, the children are not only accessing the environment and the information in it, but also



Figure 9

Frequency of social behaviors for Gary during free play sessions



decreasing a perseverative behavior (Charlop & Milstein, 1989).

In conclusion, the present study extends the literature by providing additional adjustments, analyses, and dependent variables when incorporating video modeling techniques into intervention strategies for children with autism. The results suggest that video modeling can be an effective technology for teaching children with autism to vary their conversational speech in order to speak with several persons on a variety of topics.

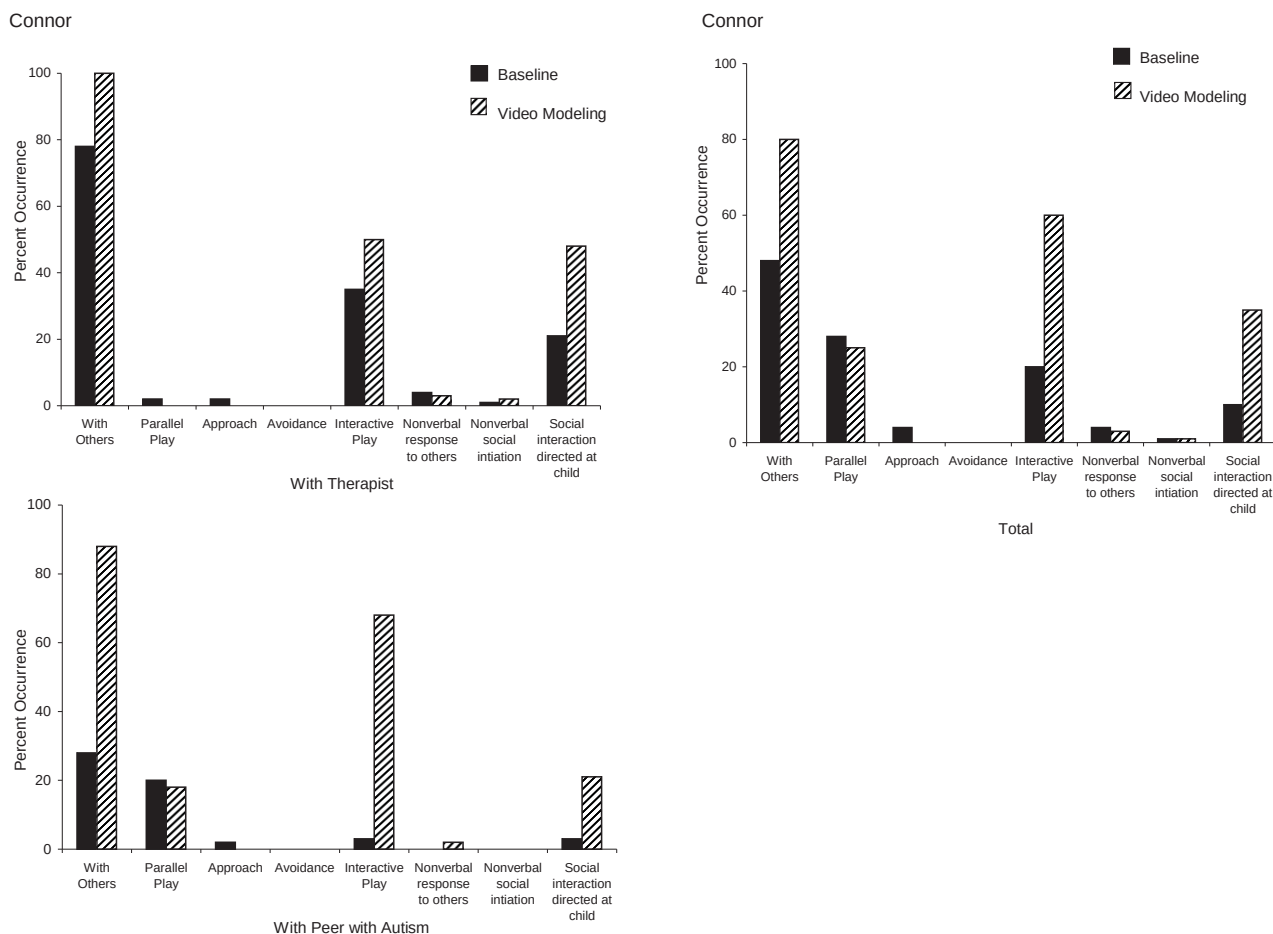
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Figure 10

Frequency of social behaviors for Connor during free play sessions



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